

The Playful AI Cookbook

Playful Programming and Artificial Intelligence · University of Copenhagen

What you are making

One recipe for a shared cookbook.

Together the group produces a collection of playful AI activities, published as a website, that any gymnasium teacher can pick up and adapt. Your recipe is your contribution: one activity, rooted in your own subject, written so that a colleague who never attended this course could run it.

A recipe is not a lesson plan. It is an idea, described clearly enough to be cooked, with honest notes about what goes wrong.

Why a cookbook

Three reasons, and they shape how you should write:

- 1 To share.** The audience is other teachers, not us. Write for someone with your subject knowledge and none of your workshop experience.
- 2 To get students talking about AI.** The best recipes give students something to argue about, not just something to operate.
- 3 To fit real subjects.** AI activities that float free of any subject get taught once and forgotten. Yours should belong somewhere on your syllabus.

Format

Document, one file, roughly 600–1000 words plus any sketches or screenshots. Use the template below. Keep the headings; they become the site structure.

Template

EIGHT SECTIONS · KEEP THE HEADINGS

01 Domain

ONE PARAGRAPH

Subject, level, and where this sits on the syllabus.

Which topic does it attach to? What are students already supposed to be learning when this appears?

"2.g biologi, photosynthesis unit, after the light-dependent reactions" beats "science, upper secondary".

Be specific.

02 The learning activity

VERBS, NOT TOPICS

What students actually do, start to finish.

- How long does it take?
- What is the sequence of steps?
- What do students produce, decide, or argue about?
- What is the one concept this carries? (One. Not four.)

Write it as instructions someone could follow.

03 The sketch

ROUGH IS FINE

The technology idea, roughed out.

This is a sketch, not a finished tool: a p5.js idea, a prompt design, an unplugged mechanic, a Teachable Machine setup, a paper prototype. Describe it well enough that someone could build a rough version in an hour. Include whatever makes it real: a diagram, a screenshot, a sample prompt, a few lines of code, a photo of the paper version.

Say what exists and what is still imagined. Both are fine.

04 What could go wrong

AT LEAST TWO

At least two known failure modes, with what to do about them.

The technical ones matter (the API is slow, the webcam fails) but the pedagogical ones matter more: where do students get confused, bored, or stuck on the wrong thing?

05 Evaluation plan

THREE OR FOUR SENTENCES

How would you know if it worked? Keep this light.

- What would you look for while students are doing it?
- What would you ask them afterwards? (One or two questions.)
- What would tell you it did NOT work?

You are not designing a study. You are deciding in advance what evidence you would accept.

06 AI Fluency

NAME ONE OR TWO

Which of the four competencies does this activity develop? Name one or two, not all four, and say how.

- Delegation — deciding whether, when and how to involve AI
- Description — communicating clearly to get useful output
- Discernment — judging whether the output is any good
- Diligence — taking responsibility and being transparent

Then the honest question: what does this activity simplify, and what could a student wrongly conclude from it?

07 Reflection

AROUND 300 WORDS

Use the Gibbs cycle as scaffolding, not as headings to fill in mechanically.

- Description. What did you actually do — designing it, and trying it out on colleagues or students?
 - Feelings. What was it like? Where were you uncertain, frustrated, or surprised?
 - Evaluation and analysis. What worked and what did not, and why do you think so? Use the two lenses from the course — design thinking (where did you go wide, where did you commit, and did you commit too early? Which phase did you skip? What did the playtest tell you that your own judgement did not?) and AI Fluency (did the activity develop the competency you claimed? Which of the four turned out to matter most in practice, and was it the one you designed for?).
 - Conclusion. What do you now understand about designing playful AI activities that you did not in August?
 - Action plan. What will you do differently next time, concretely enough that you could start on Monday?
-

08 Credits and remix notes

SHORT

What you borrowed and from whom, including from this course and from each other.

What you would try next if you ran it again.

What makes a good recipe

Specific beats general.

"Students prompt an AI to explain a concept" is not a recipe. "Students write three prompts to make the model explain diffusion to a sceptical 8-year-old, then vote on which output they would actually hand to a younger sibling" is a recipe.

Honest about the cut.

Every activity simplifies something. Saying what you cut is not a weakness in the recipe, it is the most useful part for the next teacher.

Something to argue about.

The strongest activities produce disagreement among students, not just output. Look for the moment where students have to take a position.

Runnable by someone else.

The test is not whether you could run it. It is whether a colleague could.

A NOTE ON THE REFLECTION

The reflection is for you and for the next teacher, in that order. It is the only part of the recipe that records what the design cost — the false starts, the moment you realised the activity was teaching the wrong thing, the playtest that went sideways.

Two things worth being honest about, because they are the hardest to see from outside:

- **Where design thinking helped and where it did not.** Going wide is uncomfortable when you already like your first idea. Say if you skipped it.
- **Whether the AI Fluency competency you designed for is the one the activity actually developed.** These often diverge. An activity built to teach Description frequently turns out to be mostly about Discernment, because students spend their time judging output rather than crafting input. That divergence is a finding, not a failure.

Keep it short. A tight 300 words that names one real difficulty beats a page of general satisfaction.

Length and effort

Aim for something you could genuinely write in two focused hours. This is a recipe card, not a paper. Short and clear beats long and thorough.

Timeline

WHEN	WHAT
End of the August block	Recipe drafted, at least in outline
Between August and October	Further sketch the ideas out, test on colleagues and students
October, Day 4	We will experience each other's sketches and refine the ideas
October, Day 5	Some more quick feedback, and the making of the online cookbook

Write the reflection last, after Day 4. It needs the playtest to have happened.

ONE LAST THING

You are writing for a teacher you have not met, who has fifteen minutes to decide whether to try your idea on Thursday.

Write for them.